

Dimension Reduction and Principal Component Regression

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Outline

Why Dimension Reduction?

Principal Component Analysis

PCA in Genomics

Principal Component Regression

Review

The High-Dimensional Data Problem

- ▶ Throughout this week we have worked with genomics data where $p \gg n$: tens of thousands of genes and only a handful of samples.
- ▶ But dimension reduction is not only a genomics problem. It appears any time you have many correlated variables:
 - ▶ Dozens of health indicators from a clinical exam (blood pressure, cholesterol, BMI, pulse, sleep, etc.)
 - ▶ Hundreds of survey items measuring similar constructs
 - ▶ Environmental exposures measured at many locations and times
- ▶ Three related problems:
 1. **Visualization.** You cannot plot 20 variables at once.
 2. **Collinearity.** Correlated predictors destabilize regression coefficients.
 3. **Overfitting.** More predictors than observations means a model that memorizes noise.
- ▶ Dimension reduction creates a smaller set of new variables that summarize the original ones, addressing all three.

Two Kinds of Dimension Reduction

- ▶ **Feature selection:** keep a subset of the original variables and discard the rest.
 - ▶ Example: LASSO sets many coefficients to exactly zero.
 - ▶ The surviving variables retain their original meaning.
- ▶ **Feature extraction:** transform the original variables into a new, smaller set.
 - ▶ The new variables are combinations of the originals.
 - ▶ Example: PCA creates weighted sums of all variables.
 - ▶ Individual variables lose their original meaning; patterns across variables become visible.
- ▶ Today: feature extraction, specifically PCA and how to use it in regression.

Global vs. Local Methods

- ▶ A useful distinction for knowing which tool to reach for:

Goal	Type	Method
Regression, prediction, inference	Global	PCA, Factor Analysis
Exploratory visualization	Local	t-SNE, UMAP

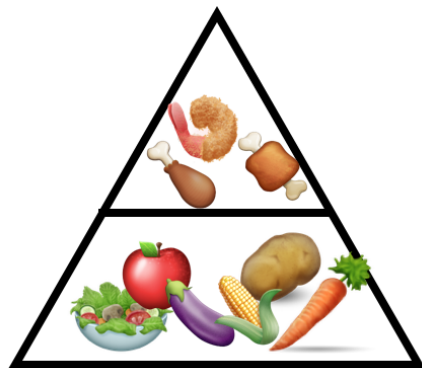
- ▶ **Global methods** preserve the overall variance structure. The transformation is linear and deterministic: you can apply it to new data, interpret coefficients on the new variables, and project new observations the same way.
- ▶ **Local methods** preserve neighborhood relationships between nearby points. The axes have no fixed meaning. You cannot easily project new data or interpret regression coefficients on local embeddings.
- ▶ Today we focus on PCA – a global method – because it is what you need for regression.

What Is PCA?

- ▶ PCA creates a new set of variables called **principal components** (PCs).
- ▶ Each PC is a weighted combination of the original variables.
- ▶ The PCs are designed to satisfy two properties:
 1. They are **uncorrelated** with each other.
 2. They are **ordered by variance explained**: PC1 captures the most variation, PC2 the next most, and so on.
- ▶ Think of it as rotating the coordinate system to align with the directions of greatest spread in the data.
- ▶ In R : `prcomp(X, center = TRUE, scale. = TRUE)`

Unsupervised Learning Example

- ▶ Consider data from the United States Department of Agriculture.
- ▶ This data contains the nutritional content of one serving of several food items (Parsley, Kale, Broccoli, Corn, Chicken, Beef, etc.).

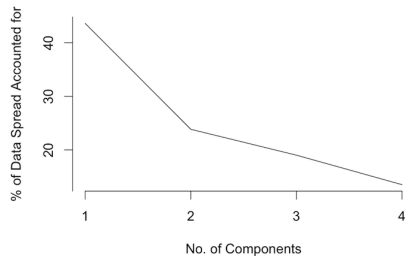


PCA Output: Three Things

- ▶ **Loadings** (the rotation matrix): the weight each original variable contributes to each PC. A large loading (positive or negative) means that variable is influential in that component. Large loadings help you interpret what a PC represents.
- ▶ **Scores** (the projected data): each observation's value on each PC. These are what you use downstream in regression, clustering, or visualization.
- ▶ **Eigenvalues / variance explained**: how much of the total variance each PC captures. Used to decide how many PCs to keep.

How Many Components to Keep?

- ▶ Keep the smallest number of PCs that explain most of the variation.
- ▶ **Scree plot:** plot eigenvalues (variance explained) vs. component number. Look for an “elbow” where the curve flattens.
- ▶ **Cumulative variance:** keep enough PCs to explain 80–95% of total variance. The right threshold depends on the application.
- ▶ **Kaiser’s rule:** keep PCs with eigenvalue > 1 (when using the correlation matrix). Simple and widely used.
- ▶ There is no single correct answer.



Interpreting Loadings

- ▶ Each loading is the correlation between the original variable and the PC (when the data are standardized).
- ▶ **Large positive loading:** the variable increases as the PC increases.
- ▶ **Large negative loading:** the variable decreases as the PC increases.
- ▶ **Near-zero loading:** the variable contributes little to this PC.
- ▶ **Example (NHANES health indicators):**
 - ▶ PC1 might have large loadings on age, systolic BP, and total cholesterol – a “cardiovascular aging” component.
 - ▶ PC2 might have large loadings on BMI and diastolic BP – a “metabolic risk” component.
- ▶ The sign of the loadings is arbitrary. Only the relative magnitude and pattern across variables matter.

	PC1	PC2	PC3	PC4
Fat	-0.45	0.66	0.58	0.18
Protein	-0.55	0.21	-0.46	-0.67
Fiber	0.55	0.19	0.43	-0.69
Vitamin C	0.44	0.70	-0.52	0.22

Figure: Factor loadings for the food example.

The Biplot

- ▶ A **biplot** shows both observations (points) and variables (arrows) in the same plot.
- ▶ Each arrow points in the direction the original variable increases. Arrow length indicates how much variance the variable contributes to the displayed PCs.
- ▶ Two observations with similar scores on both PCs lie close together.
- ▶ Two variables whose arrows point in the same direction are positively correlated in the data; arrows pointing opposite ways indicate negative correlation.
- ▶ In R : `fviz_pca_biplot(pca_res)` from the `factoextra` package.

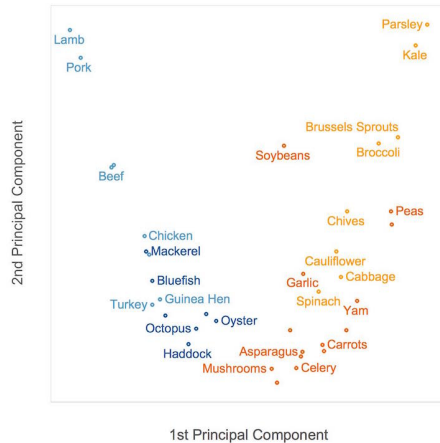


Figure: First and second PCs for the food example.

Important Practical Notes on PCA

- ▶ **Scale your variables first.** PCA uses variance. Variables measured in large units (income in dollars) will dominate variables measured in small units (blood pressure in mmHg) unless you standardize. Use `scale. = TRUE` in `prcomp()`.
- ▶ **PCA is sensitive to outliers.** A single extreme observation can pull a PC toward itself. Check for outliers in the score plot before trusting the components.
- ▶ **PCA is unsupervised.** It finds the directions of maximum variance in the predictors X , regardless of the outcome Y . There is no guarantee that the directions of maximum variance are also the directions most predictive of Y .
- ▶ **You can project new data.** Because PCA is a linear transformation, you can apply the same rotation to new observations: `predict(pca_res, newdata = X_new_scaled)`.

PCA in the Genomics Context

- ▶ You have already used PCA this week – the `plotPCA()` call on Day 11 and the VST PCA in Day 13 were exactly `prcomp()` applied to the gene expression matrix.
- ▶ In genomics, PCA serves three main purposes:
 1. **Quality control.** Do samples cluster by treatment group? Are there outliers or batch effects? (Day 11)
 2. **Visualization.** Reduce 20,000 genes to 2 dimensions for a readable plot. (Days 11–13)
 3. **Covariate adjustment.** The top PCs can be included as covariates in a regression model to remove global confounders (population stratification in GWAS, batch effects in RNA-seq).
- ▶ The genomics PCA uses the VST-transformed count matrix as X . Each sample is one observation; each gene is one variable.

The Collinearity Problem

- ▶ Recall from Day 6: when predictors are highly correlated in linear (or logistic) regression, coefficient estimates become unstable.
 - ▶ Standard errors blow up.
 - ▶ Coefficient signs can flip when a correlated predictor is added or removed.
 - ▶ The model is sensitive to small changes in the data.
- ▶ Example: predicting blood pressure from a clinical panel that includes BMI, weight, waist circumference, and cholesterol. All are highly correlated. No single coefficient is interpretable.
- ▶ PCA solves this by replacing the correlated predictors with a smaller set of uncorrelated components.

Principal Component Regression (PCR)

► **PCR in two steps:**

1. Run PCA on the predictor matrix $\mathbf{X}$. Keep the first r principal components.
2. Regress the outcome Y on the r PC scores instead of the original predictors.

- The PC scores are uncorrelated by construction, so collinearity is eliminated completely.
- The regression coefficients on the PCs are interpretable: $\hat{\gamma}_k$ is the change in Y for a one-unit increase in PC k .
- In R :
- `pca_res <- prcomp(X, scale. = TRUE)`
 - `scores <- as.data.frame(pca_res$x[, 1:r])`
 - `lm(Y ~ ., data = cbind(Y, scores))`

How Many PCs for Regression?

- ▶ For quality control and visualization, you keep the PCs that explain the most variance in X .
- ▶ For regression, the “right” number of PCs is the one that gives the best predictive performance, not necessarily the one that explains the most variance in X .
- ▶ These two criteria can disagree. A PC that explains little variance in X might be highly associated with Y , and a PC that explains a lot of variance in X might be irrelevant to Y .
- ▶ In practice: use cross-validation to choose r .
 - ▶ Fit the PCR model for $r = 1, 2, \dots, p$.
 - ▶ Pick the r that minimizes cross-validated prediction error.
 - ▶ In R : the `pls` package provides `pcr()` with built-in cross-validation.

PCR: Advantages and Limitations

Advantages	Limitations
Eliminates collinearity	PCs may not align with Y
Reduces overfitting	Interpretability is harder
Works when $p > n$	Number of PCs is a tuning choice
Applicable to new data	PCA is unsupervised (ignores Y)

- ▶ The key limitation: PCA is unsupervised. The directions of maximum variance in X are found without looking at Y . If the predictive signal lives in a direction of low variance, PCR will miss it.
- ▶ For most applied problems with correlated predictors and a modest number of components, PCR works well in practice.

Review: Key Points

- ▶ **Dimension reduction:** Global methods (PCA) preserve overall variance structure and can be used in regression. Local methods (t-SNE, UMAP) preserve neighborhood structure and are for visualization only.
- ▶ **PCA** creates uncorrelated components ordered by variance explained.
 - ▶ PCA Output: loadings (interpret the components), scores (use in regression or plots), eigenvalues (choose how many to keep).
 - ▶ Always scale variables first. Check for outliers. Use a scree plot or cumulative variance to choose the number of components.
- ▶ **PCR:** Replace correlated predictors with uncorrelated PCA scores, then regress Y on the scores. Eliminates collinearity; choose the number of PCs by cross-validation.